

Test Plan – Shapes Calculator

1. Objective

To verify that the shape calculations (perimeter & area) work correctly for different shape types (Square, Rectangle, Triangle).

2. Scope

Testing of CalculatePerimeter() & CalculateArea() methods for each different shape with different lengths.

3. Test Items

- Square
- Rectangle
- Triangle

4. Test Approach

Unit testing using predefined inputs and expected outputs, along with Assert.Fail() where appropriate.

5. Test Cases

1. Square Perimeter Calculation

Input: [5]

Expected Output: 20

2. Rectangle Perimeter Calculation

Input: [5, 3]

Expected Output: 16

3. Triangle Perimeter Calculation

Input: [10, 10, 10]

Expected Output: 30

4. Square Area Test

Input: [5]

Expected Output: 25

5. Rectangle Area Test
Input: [5, 3]
Expected Output: 15
6. Triangle Area Test
Input: [10, 10, 10]
Expected Output: 43
7. Square Input Equivalence Partition Test
Input: [5, 5]
Expected Output: ArgumentException
8. Square Negative Length Equivalence Partition Test
Input: [-5]
Expected Output: ArgumentException
9. Rectangle Input Equivalence Partition Test
Input: [5, 5, 5]
Expected Output: ArgumentException
10. Rectangle Negative Length Equivalence Partition Test
Input: [-5, 3]
Expected Output: ArgumentException
11. Triangle Input Equivalence Partition Test
Input: [10, 10, 10, 10]
Expected Output: ArgumentException
12. Triangle Negative Length Equivalence Partition Test
Input: [-5, 5, 5]
Expected Output: ArgumentException
13. Triangle Values Equivalence Partition Test
Input: [1, 2, 3]
Expected Output: ArgumentException
14. Square Stress Test
Input: [10000]
Expected Output: 100000000
15. Rectangle Stress Test
Input: [20000, 10000]
Expected Output: 200000000

16. Triangle Stress Test

Input: [20000, 20000, 10000]

Expected Output: 96824583

17. Square Boundary Test

Input: [0]

Expected Output: ArgumentException

18. Rectangle Boundary Test

Input: [0, 0]

Expected Output: ArgumentException

19. Triangle Boundary Test

Input: [0, 0, 0]

Expected Output: ArgumentException

6. Expected Result

All test cases should pass successfully.